Division of Applied Mathematics

Brown University

Subject : APMA 2070-Deep Learning for Scientists and Engineers Year : Spring 2026

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Homework : 5 Due Date : 04/11/2026

Instructions

This homework compares **polynomial interpolation**, **Fourier series**, and **neural networks** as *basis expansions*, emphasizing **stability, conditioning, and failure modes**.

— You may use AI tools for syntax, debugging, or plotting.

— All explanations, diagnostics, and interpretations must be your own.

— You are responsible for understanding and defending every result you submit.

Unless otherwise stated, use double precision throughout and clearly report : (i) grids/sampling strategy, (ii) model architecture/training budget, and (iii) error metrics.

A brief review on function approximation and polynomials representation

The **Weierstrass theorem** states that *any continuous function defined in a finite range can be approximated to any degree of accuracy by polynomial powers*. This theorem does not state what is the appropriate type of polynomial approximation. The question which still remains to answer is whether equidistant data always produce stable and convergent polynomial approximations. O. Runge in 1901 demonstrated the non convergent polynomial approximation

using equidistant points for the following function

$$f(x) = \frac{1}{1 + 25x^2}, \qquad x \in [-1, 1] \tag{1}$$

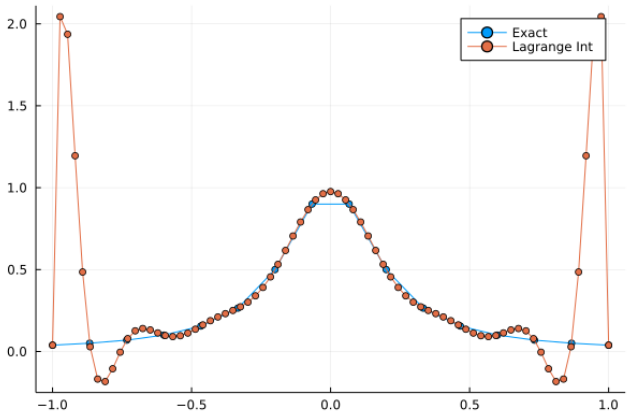

FIGURE 1 – Interpolation of function in (1) using equidistant grid.

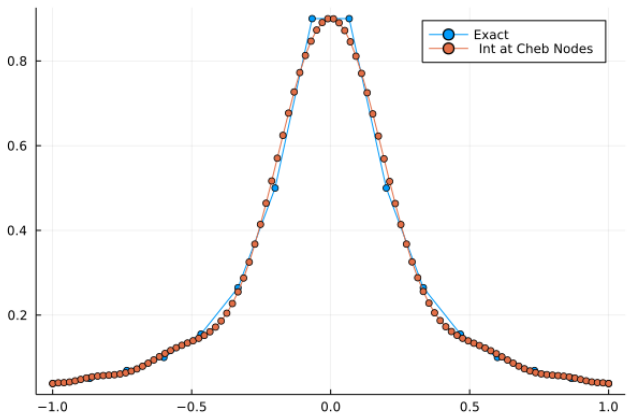

FIGURE 2 – Interpolation of function in (1) using Chebyshev gridpoints.

The Barycentric Lagrange Interpolation in 1D

Given grid x_j, we need to obtain polynomial $p(x_j)$ such that

$$p(x_j) = f(x_j) \qquad j = 0 \dots n$$

Solution to the above problem is expressed as Lagrange formulation

$$p(x_j) = \sum_{i=0}^{n} f(x_i) l_i(x), \;\; l_i(x) = \frac{\prod_{k=0,k\neq j}^{n}(x - x_k)}{\prod_{k=0,k\neq j}^{n}(x_i - x_k)} \tag{2}$$

This can be written simply by using the Barycentric Lagrange form

$$p(x_j) = \frac{\sum_{j=0}^{n} \frac{w_j}{x-x_j} f(x_j)}{\sum_{j=0}^{n} \frac{w_j}{x-x_j}}, \tag{3}$$

where the barycentric weights are

$$w_j = \frac{1}{\prod_{k\neq j}(x_j - x_k)}$$

For equidistant grid

$$w_j = (-1)^j \binom{n}{j}$$

Interpolation of (1) using equidistant grid is given in Figure 1. Interpolation of (1) using Chebyshev grids is given in Figure 2. The Chebyshev grid points are expressed as

$$x_j = \cos(j\pi/N).$$

Function approximation using Fourier Series and Deep Neural Network

A periodic function $f(x)$ can be written as linear combination of cosine and sine of different frequency, which reads

$$f(x) = \frac{A_0}{2} + \sum_{k=1}^{\infty} A_k \cos(kx) + B_k \sin(kx), \tag{4}$$

where

$$A_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(kx)dx = \frac{1}{||\cos(kx)||^2}\langle f(x), \cos(kx)\rangle$$

$$B_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(kx)dx = \frac{1}{||\sin(kx)||^2}\langle f(x), \sin(kx)\rangle$$

Now, we will express (4) for any finite domain between $[0, L]$

$$f(x) = \sum_{k=0}^{\infty}\left(A_k \cos\left(\frac{2\pi kx}{L}\right) + B_k \sin\left(\frac{2\pi kx}{L}\right)\right) \tag{5}$$

where

$$A_k = \left\langle f(x), \cos\left(\frac{2\pi kx}{L}\right)\right\rangle$$

$$B_k = \left\langle f(x), \sin\left(\frac{2\pi kx}{L}\right)\right\rangle$$

Example and implementation of function approximation using Fourier series

Consider a hat function expressed as

$$f(x) = \begin{cases} \frac{\pi}{2} - |x|, & |x| < \frac{\pi}{2} \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

The plot of (6) is shown in Figure 3. The approximation of (6) using (5) is shown in Figure 4.

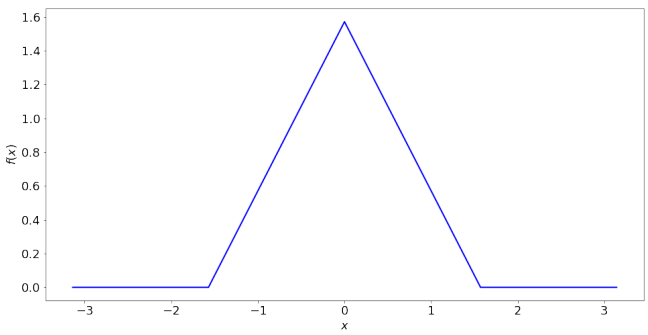

FIGURE 3 – Hat function

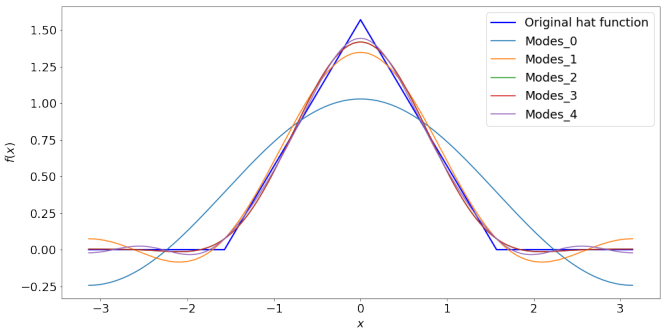

FIGURE 4 – Approximation of hat function using 5 modes.

Task 1 : Polynomial Interpolation, Stability, and the Runge Phenomenon

Consider Runge's function expressed by (1) and barycentric Lagrange interpolant (3) and perform following

(a) **Equidistant vs Chebyshev nodes.** For $N \in \{10, 20, 40\}$, construct interpolants of f using :

— equidistant nodes on $[-1, 1]$,

— Chebyshev–Lobatto nodes $x_j = \cos(j\pi/N)$ mapped to $[-1, 1]$.

Evaluate errors on a dense test grid (at least 2000 points). Plot the pointwise error $|f(x) - p(x)|$ on a **log scale**. Identify the spatial location(s) of the largest errors.

(b) **Conditioning via barycentric weights.** For each node set and each N, compute the barycentric weights $\{w_j\}$ and report :

— $\max_j |w_j|$,

— $\text{range}(|w|) = \max_j |w_j| / \min_j |w_j|$ (excluding zero),

— a brief explanation of how weight magnitudes relate to numerical stability.

(Do not just state "Chebyshev is better" : connect your observation to amplification of perturbations.)

(c) **Perturbation sensitivity test.** Add i.i.d. noise of size 10^{-10} to the data values $f(x_j)$ and recompute the interpolant. For each node set :

— plot the perturbed error $|f(x) - \tilde{p}(x)|$ (log scale),

— report the *amplification factor* $\|\tilde{p} - p\|_\infty / 10^{-10}$,

— explain where in the domain the perturbation is most amplified and why.

(d) **One-sentence takeaway.** In one sentence : what is the main reason equidistant interpolation can fail for (1) ?

Task 2 : Fourier Series, Convergence, and Gibbs Phenomenon

Consider the "hat" function (6), defined on $[-\pi, \pi]$ and let $S_K(x)$ denote the truncated Fourier series approximation with K modes.

(a) **Convergence diagnostics.** Compute S_K for $K \in \{5, 10, 20, 40\}$ and evaluate on a dense grid (at least 4000 points). Plot :

— pointwise error $|f(x) - S_K(x)|$,

— L^2 error versus K on a log–log scale.

(b) **Quantify Gibbs overshoot.** Near $x = \pm\pi/2$, compute the maximum overshoot :

$$\delta_K = \max_{x \in [-\pi, \pi]} (S_K(x) - f(x)).$$

Report δ_K for each K and explain why increasing K reduces oscillation *width* but not the overshoot *height* (qualitatively, no need to prove).

(c) **Localization vs global bases.** Explain (clearly, in 5–8 lines) why Fourier series can perform poorly for localized non-smooth features, even though they converge spectrally for smooth periodic functions.

Task 3 : Neural Networks as Adaptive Basis Expansions

You will approximate both (1) and (6) using fully-connected neural networks.

(a) **Controlled architectures.** Choose two network architectures with approximately the *same number of parameters* (e.g., one shallow-wide and one deeper-narrow). Use the same optimizer and a fixed training budget (number of steps/epochs). State all training details explicitly.

(b) **Sampling strategies.** For each target function, train using each of the following sampling strategies for training inputs :

— equidistant sampling,

— Chebyshev-like sampling (cluster near boundaries on $[-1, 1]$ or near $[-\pi, \pi]$ endpoints),

— random uniform sampling.

Use the same number of training points across the three strategies.

(c) **Error localization.** For each case, plot the pointwise error and identify whether the NN approximation error is (i) boundary-dominated, (ii) localized near nonsmooth regions, or (iii) globally oscillatory. Explain what this suggests about the NN acting like a polynomial basis, a Fourier basis, or an adaptive localized basis.

(d) **Interpolation vs learning.** Explain (briefly) why sampling affects NN training *differently* than polynomial interpolation. Your answer must reference at least one of : optimization, implicit regularization, or loss landscape.

Task 4 : Optimizers and Implicit Regularization

Consider following two functions

$$y(x) = \begin{cases} 5 + \sum_{k=1}^{6} \sin(kx), & x < 0, \\ \cos(10x), & x \geq 0, \end{cases} \tag{7}$$

and the step function

$$y(t) = \begin{cases} 0, & t < 1, \\ 1, & t \geq 1. \end{cases} \tag{8}$$

(a) Train a network to approximate (7) and (8) using :

— at least one first-order optimizer (e.g., Adam, SGD+momentum),

— at least one second-order or quasi-Newton method (e.g., L-BFGS).

Keep architecture and training budget controlled across optimizers as much as possible.

(b) **Failure-first analysis.** Identify at least one case where optimization fails despite seemingly sufficient network capacity. Classify the failure as primarily due to :

— nonsmoothness,

— poor conditioning,

— optimizer implicit bias,

— or optimization pathology (e.g., stagnation, exploding gradients).

Justify your classification using evidence (loss curves, error localization, or sensitivity to initialization).

(c) **Compare to Fourier failure modes.** Explain in 5–8 lines how optimizer-induced failures differ from Fourier approximation failures for nonsmooth functions.

Mandatory Reflection

Answer concisely (total $\leq$ 10 lines) :

1. Identify one approximation result that contradicted your intuition.
2. Explain the mechanism responsible (basis choice, conditioning, or optimization).
3. State one scenario where you would prefer *polynomial interpolation*, *Fourier series*, or *neural networks*, and justify.